SISTEMA DE METROLOGIA LEAN TECH – ARQUITETURA MODULAR DO NO PRINCIPAL (CARRIER BOARD)

Diagrama de Blocos Hierarquicos com Fluxo Sinalizacao Inputs/AFE -> MCU ESP32-S3 -> Outputs/LoRa

Power_Supply

VBAT_ADC

File: 01_Power_Supply.kicad_sch

Sensors_AFE

PWR_GATE_EN

1W_DQ

I2C_SCL

I2C_SDA

File: 02_Sensors_AFE.kicad_sch

MCU_ESP32S3

VBAT_ADC

PWR_GATE_EN

1W_DQ

I2C_SCL

I2C_SDA

LORA_M0

LORA_M1

LORA_TX

LORA_RX

LORA_AUX

GPIO_LED_GRN

GPIO_LED_ORG

GPIO_LED_BLU

GPIO_BUZZER

File: 03_MCU_ESP32S3.kicad_sch

Peripherals_LoRa

LORA_M0

LORA_M1

LORA_TX

LORA_RX

LORA_AUX

GPIO_LED_GRN

GPIO_LED_ORG

GPIO_LED_BLU

GPIO_BUZZER

I2C_SCL

I2C_SDA

File: 04_Peripherals_LoRa.kicad_sch

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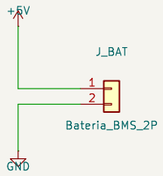
Sheet: /
File: amemiya_carrier_v2.kicad_sch

Title: Amemiya Carrier Board – Modular Architecture (Clean Baseline)

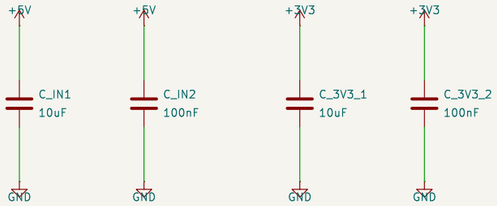
Size: A3	Date: 2026-10-16	**Rev: v36.2**
KiCad E.D.A. 9.0.7		Id: 1/5

ENTRADA DE ENERGIA 5V DO MODULO STEP-UP (BOOST)

Bateria 3.7V Li-Ion -> BMS TP4056 -> Chave ON/OFF -> Step-Up Boost 5V -> J_BAT



DESACOPLAMENTO DA ALIMENTACAO (+5V & +3V3)

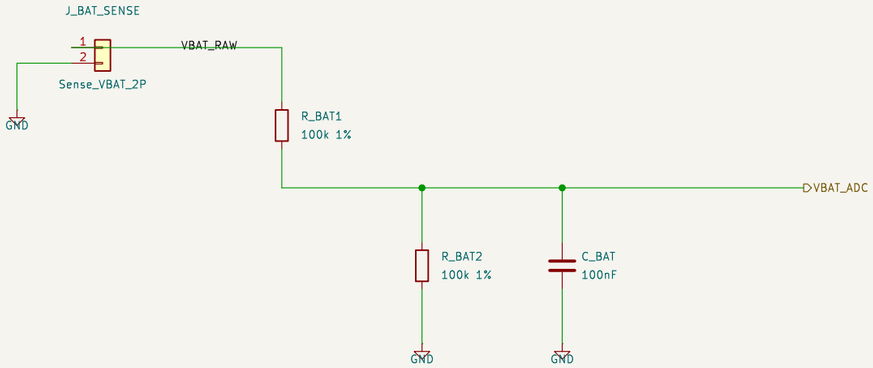


FONTES DE ALIMENTACAO DO SISTEMA (PWR_FLAG)



CIRCUITO DE MONITORAMENTO DE BATERIA (FUEL GAUGE DIVIDER)

Divisor 1:2 (100k / 100k) com filtro 100nF para leitura ADC de VBAT (0-4.2V -> 0-2.1V)



CADEIA DE ALIMENTACAO INDUSTRIAL EXTERNA (OFF-BOARD)

1. BATERIA: 2x Celulas 18650 Li-Ion (3.7V nom / 4.2V max / ~5200mAh em paralelo)

2. BMS/CARREGADOR: Modulo TP4056 com protecao contra sobrecarga e descarga profunda

3. CHAVE GERAL: Chave alavanca liga/desliga de painel (interrompe o polo positivo)

4. STEP-UP (BOOST): Modulo MT3608 ajustado para 5.00V DC regulado (eficiencia ~93%)

5. ENTRADA NA PLACA: Conector JST-XH 2P (J_BAT) -> Pino 22 do DevKit (5V IN)

6. SENSE DE BATERIA: Conector JST-XH 2P (J_BAT_SENSE) ligado antes do Step-Up

7. FORMULA FUEL GAUGE: V_ADC = V_BAT * (100k / (100k + 100k)) = V_BAT / 2 (Max 2.10V)
   -> Compativel com faixa linear do ADC1 (0 a 3.1V com atenuacao 11dB no ESP32-S3)

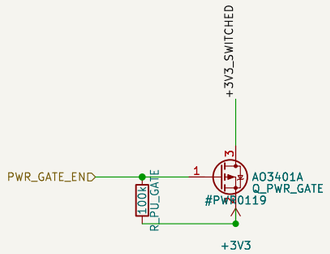
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Sheet: /Power_Supply/
File: 01_Power_Supply.kicad_sch

Title: 01 - Power Supply, Decoupling & Battery Fuel Gauge

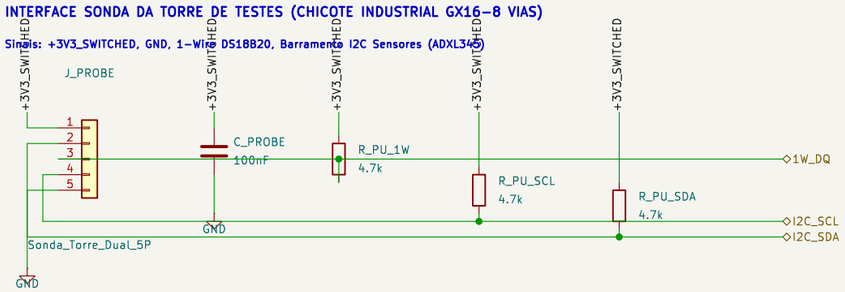
Size: A3	Date: 2026-09-26	**Rev: v36.1**
KiCad E.D.A. 9.0.7		Id: 2/5

CONDICIONAMENTO DE SINAIS (AFE) E INTERFACES DE SENSORES INDUSTRIAIS

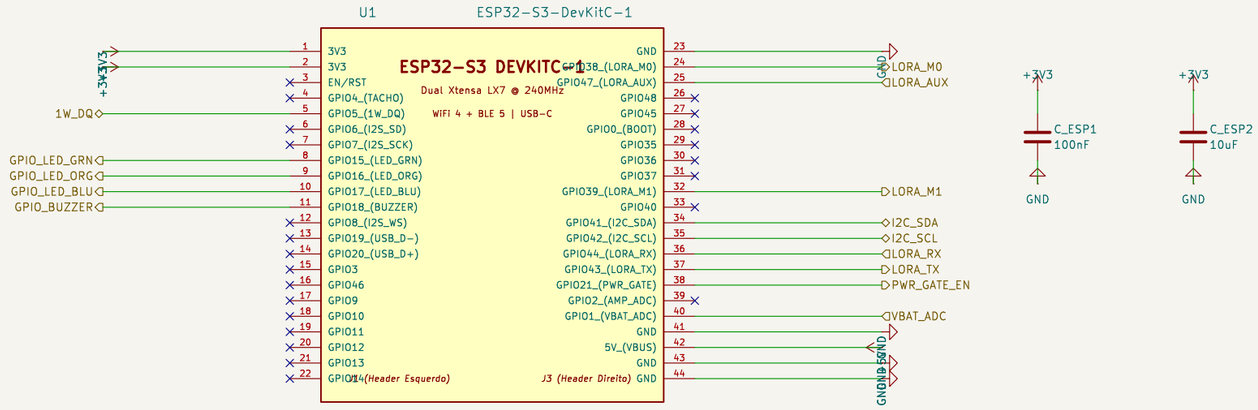


INTERFACE SONDA DA TORRE DE TESTES (CHICOTE INDUSTRIAL GX16-8 VIAS)

Sinais: +3V3_SWITCHED, GND, 1-Wire DS18B20, Barramento I2C Sensores (ADXL345)



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Sheet: /Sensors_AFE/ File: 02_Sensors_AFE.kicad_sch		
Title: 02 – Analog Front-End (AFE) & Sensor Interfaces		
Size: A3	Date: 2026-09-26	**Rev: v36.1**
KiCad E.D.A. 9.0.7		Id: 3/5

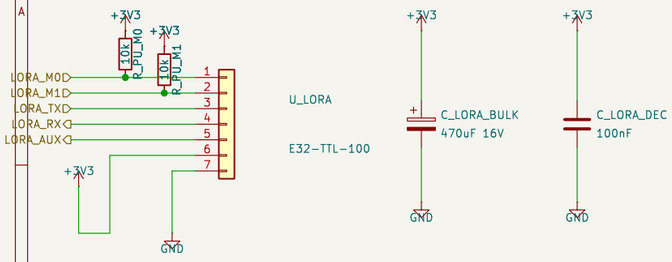


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Sheet: /MCU_ESP32S3/		
File: 03_MCU_ESP32S3.kicad_sch		
Title: 03 – ESP32-S3 DevKit 44-Pin Monolithic Interface		
Size: A3	Date: 2026-10-17	Rev: v37.0 – Single Symbol
KiCad E.D.A. 9.0.7		Id: 4/5

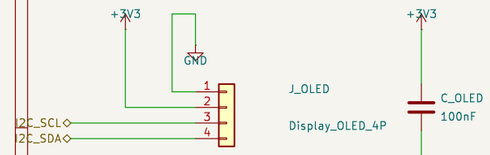
COMUNICACAO RF LORA EBYTE E32-TTL-100 / ATUADORES DE INTERFACE HUMANO-MAQUINA

TRANSCEPTOR LORA EBYTE E32-TTL-100 (433MHz / 915MHz)

Capacitor bulk 470uF Low-ESR essencial para absorver surtos de transmissao RF sem brownout

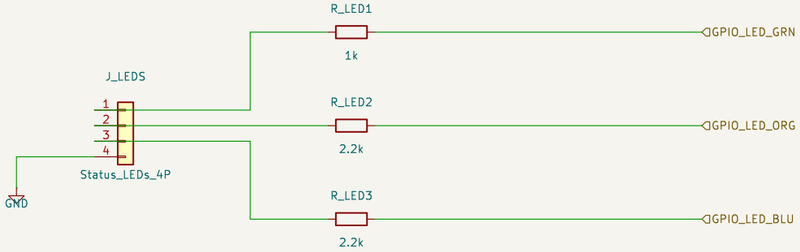


DISPLAY OLED GRAFICO 0.96 POL SSD1306 I2C

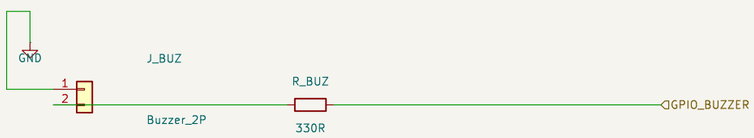




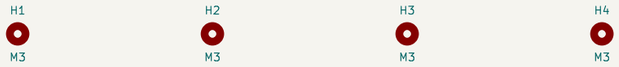
LEDS DE STATUS DO PAINEL (VERDE, LARANJA, AZUL)



BUZZER SONORO ATIVO DE ALERTA INDUSTRIAL



FUROS DE FIXACAO MECANICA (M3) NOS 4 CANTOS DA PLACA (90x65mm)



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Sheet: /Peripherals_LoRa/
File: 04_Peripherals_LoRa.kicad_sch

Title: 04 – LoRa E32 RF, OLED Display, Status LEDs & Buzzer

Size: A3	Date: 2026-09-26	Rev: v36.1
KiCad E.D.A. 9.0.7		Id: 5/5